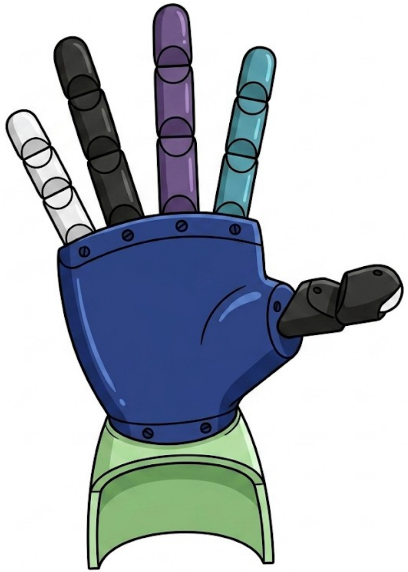
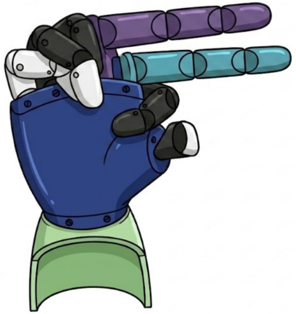


BuildBuddy @ Dartmouth

AI-Powered ASL Fingerspelling Hand Building Instructions

Designed for Dartmouth Students!



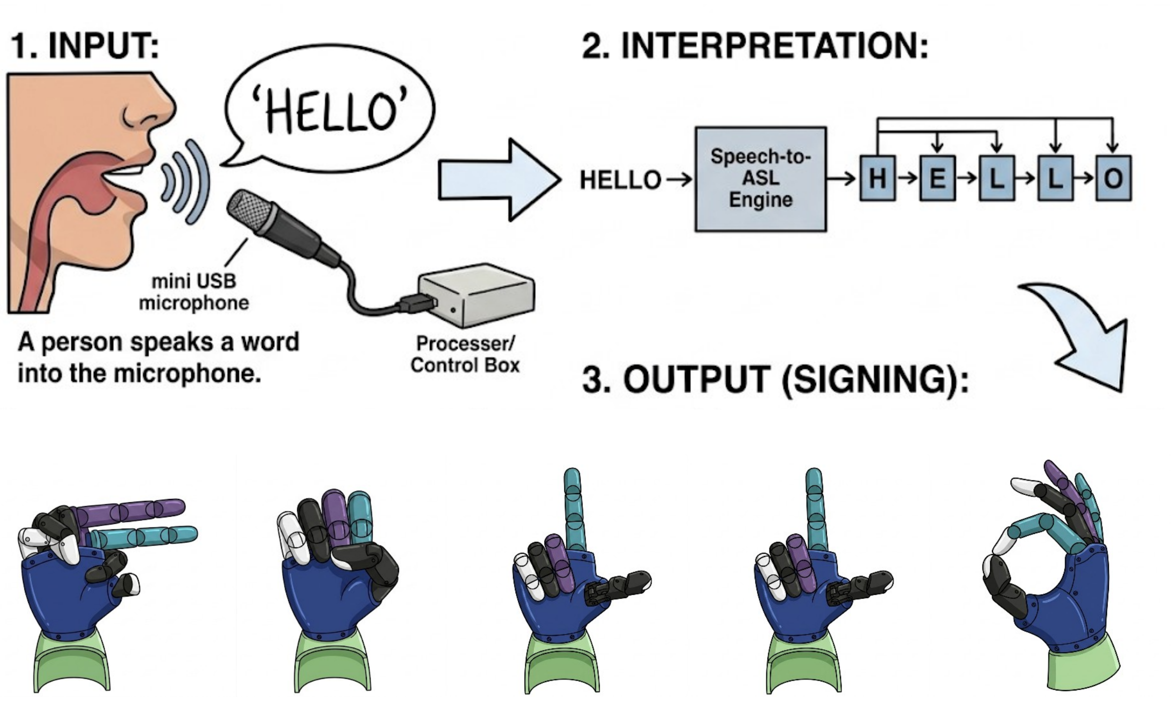
H

E

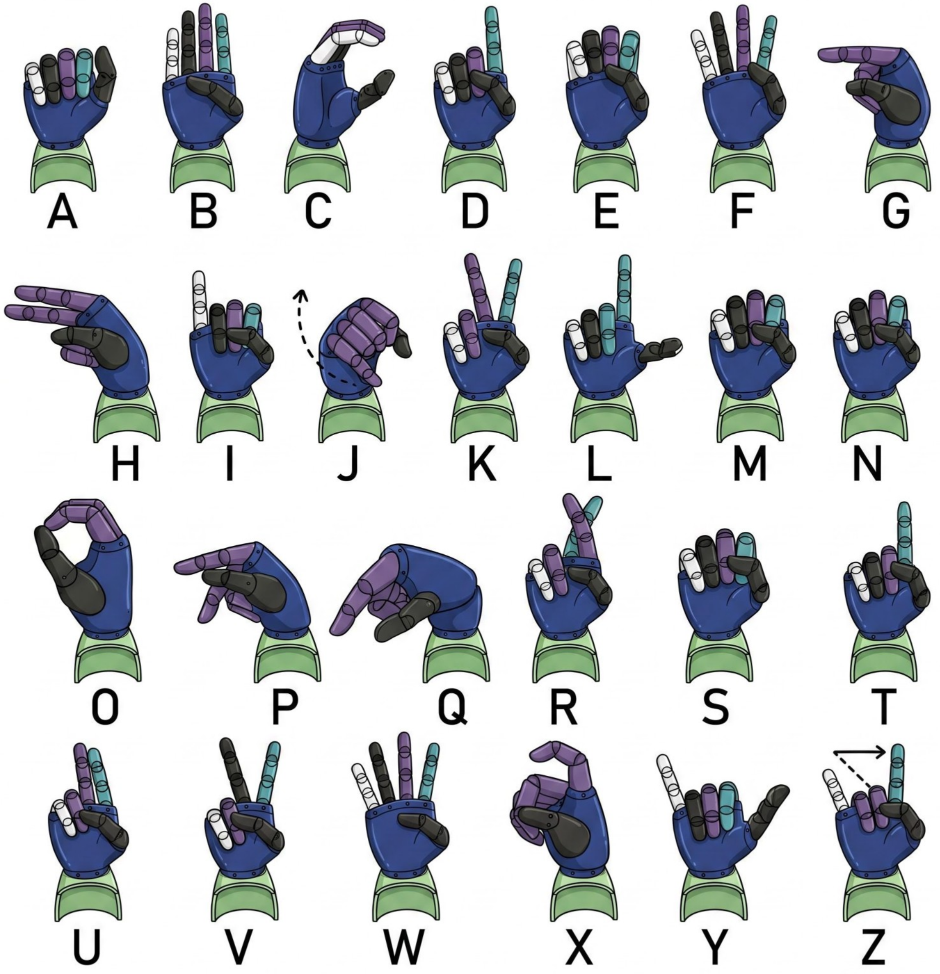
L

L

O

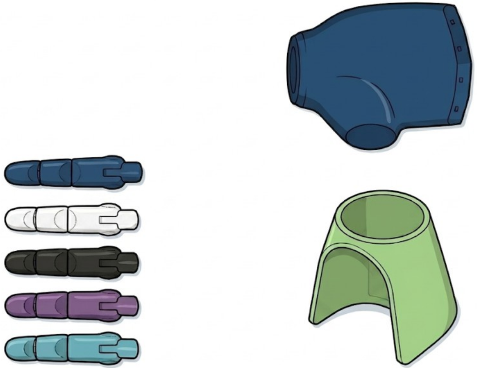


Build a 3D-printed hand that turns voice input into the ASL alphabet ***in 24 hours!***



Learn the ASL alphabet!
Learn to build!

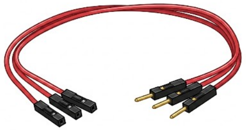
Parts Needed



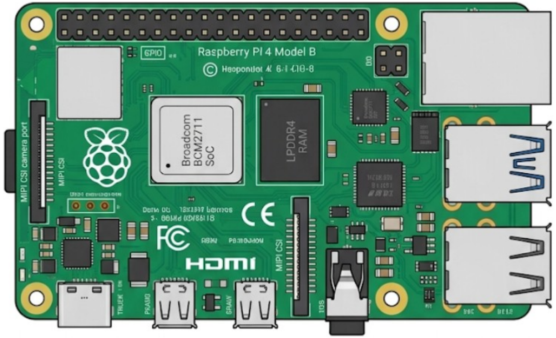
3D-printed hand parts
(see next page)

Bungee Cord

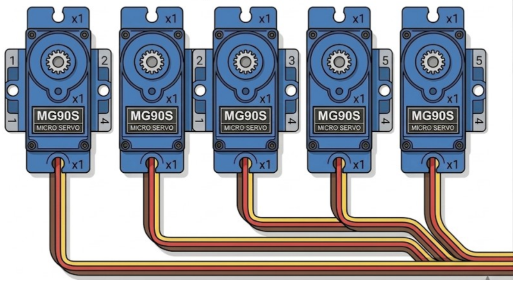
Fishing Line



Male-to-Female
Dupont Jumper Wires



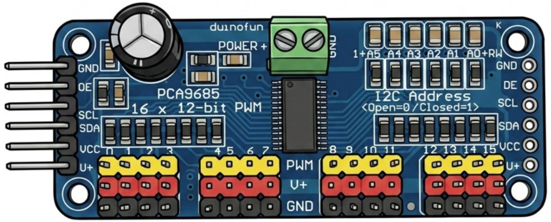
Raspberry Pi 4 Model B



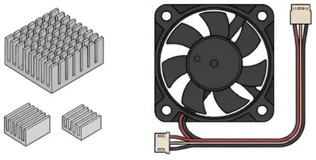
5x MG90S Micro Servos

5V 5A Power Supply
with Barrel Jack Adapter

USB
Microphone



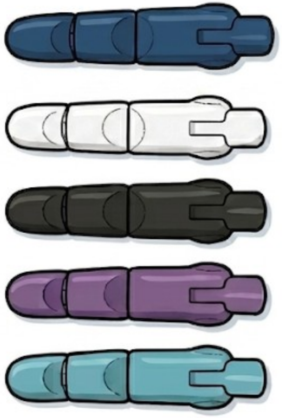
PCA9685

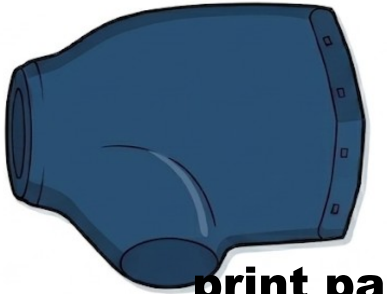


Heatsinks and Fan
for Pi

USB-C Power
Supply for Pi

3D-Printed Hand Parts





Download and print parts from https://tinyurl.com/2v6u7m3e

3D-Printing @ Dartmouth

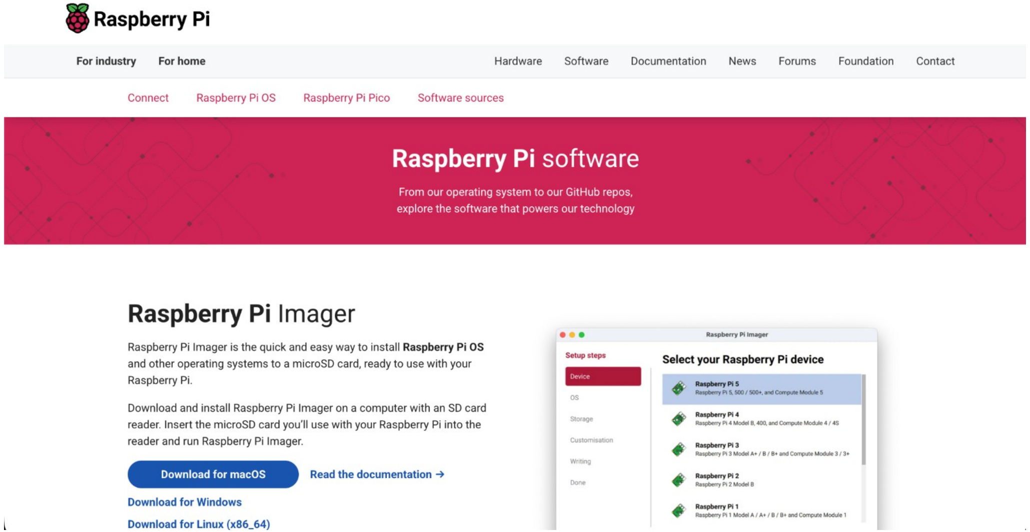
Go to MShop (C027) or Harold Edward Cable MakerSpace (ECSC 003), give the STL files from the previous page, and get them printed!

Part 

Set Up the Raspberry Pi

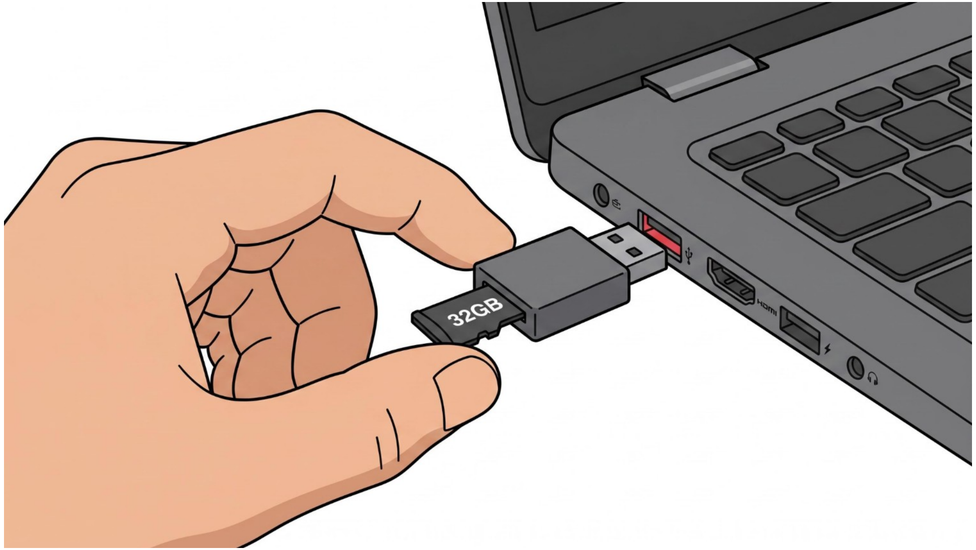
Download Raspberry Pi Imager from raspberrypi.com/software

1



Plug in your 32 GB micro SD card into your laptop (you may need a micro SD to SD card adapter)

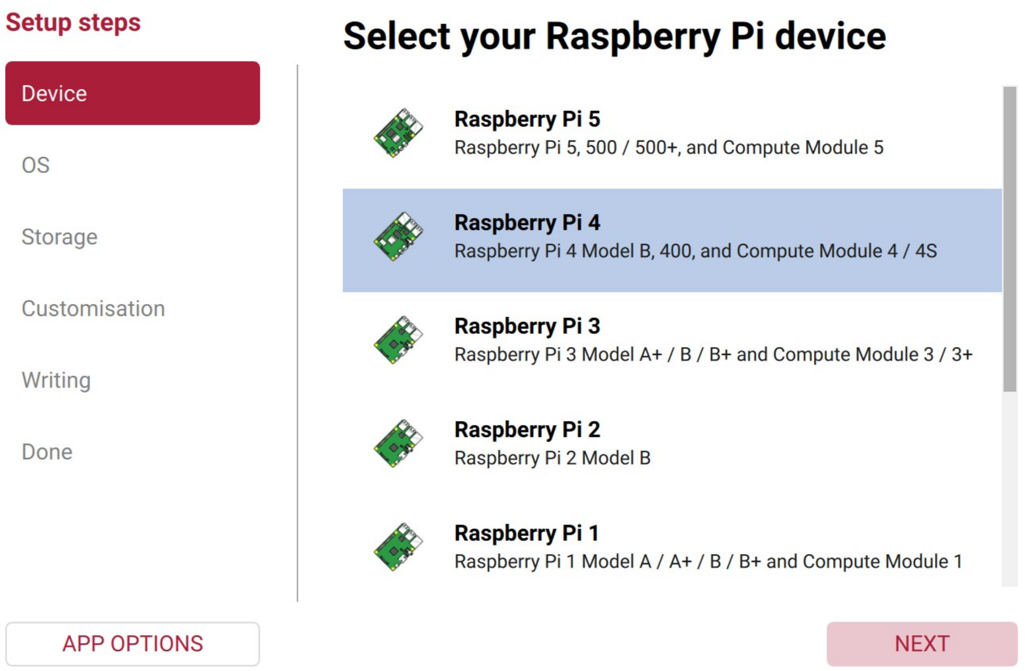
2



Open Raspberry Pi Imager

Choose Device -> Raspberry Pi 4

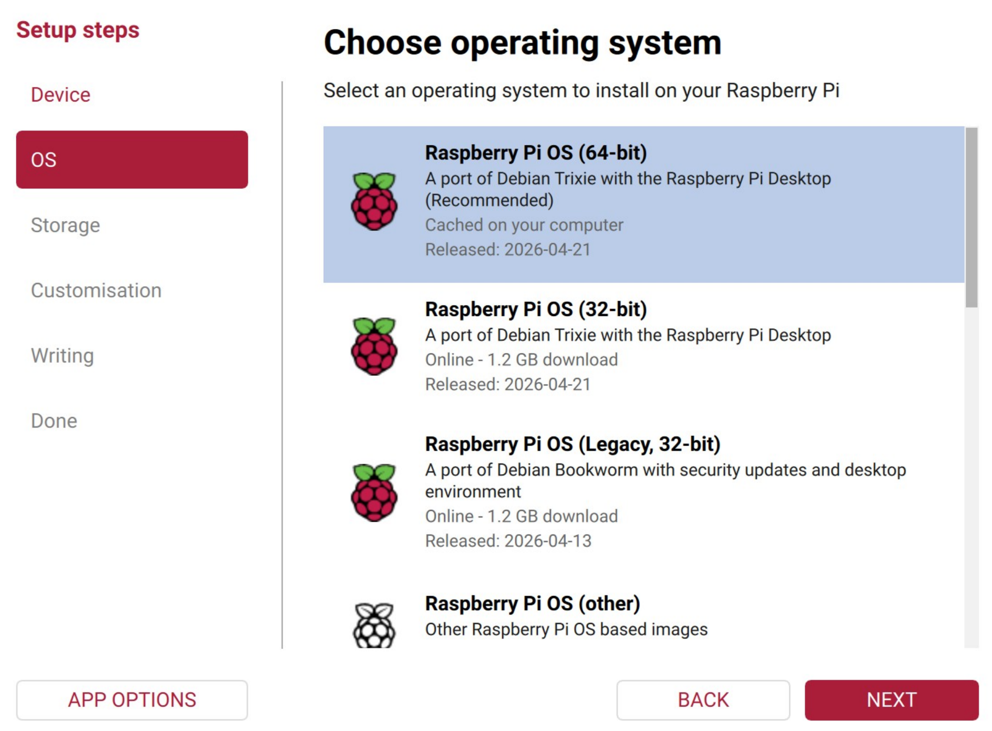
3



Open Raspberry Pi Imager

Choose OS -> Raspberry Pi OS (64-bit)

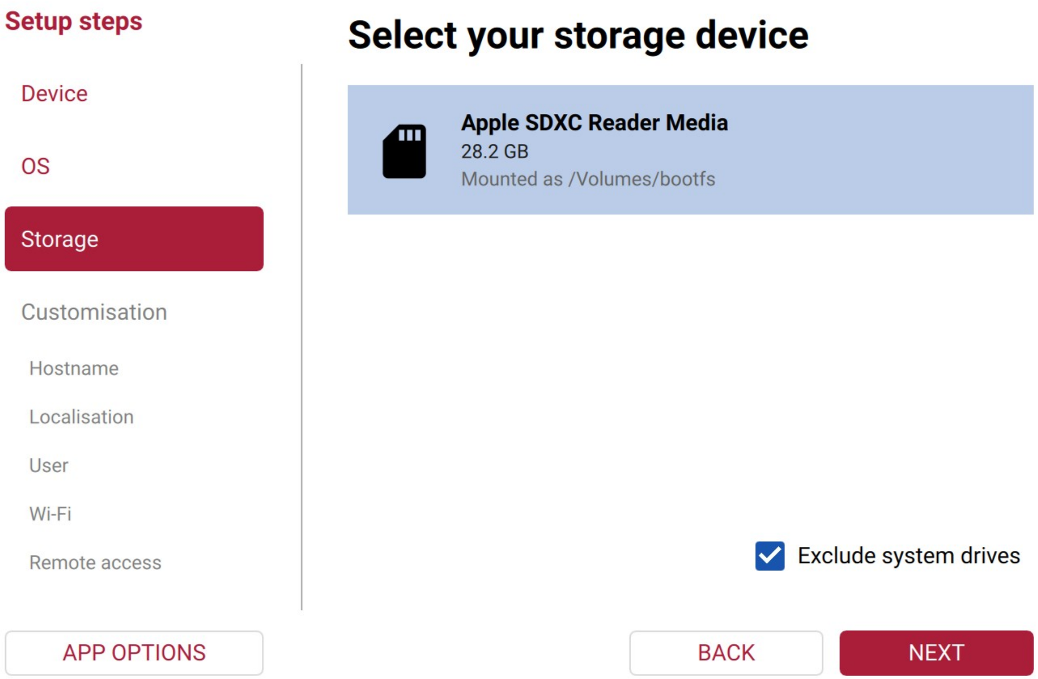
4



Open Raspberry Pi Imager

Choose Storage -> {your micro SD card}

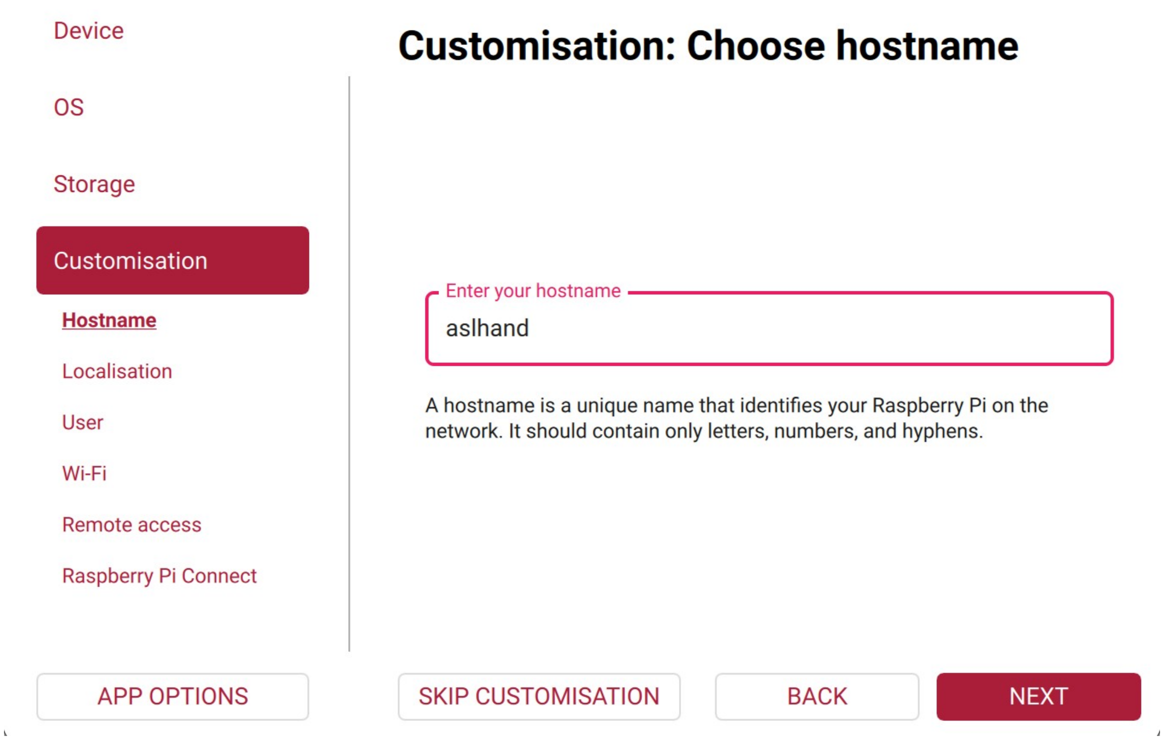
5



Open Raspberry Pi Imager

Customisation → hostname → "aslhand"

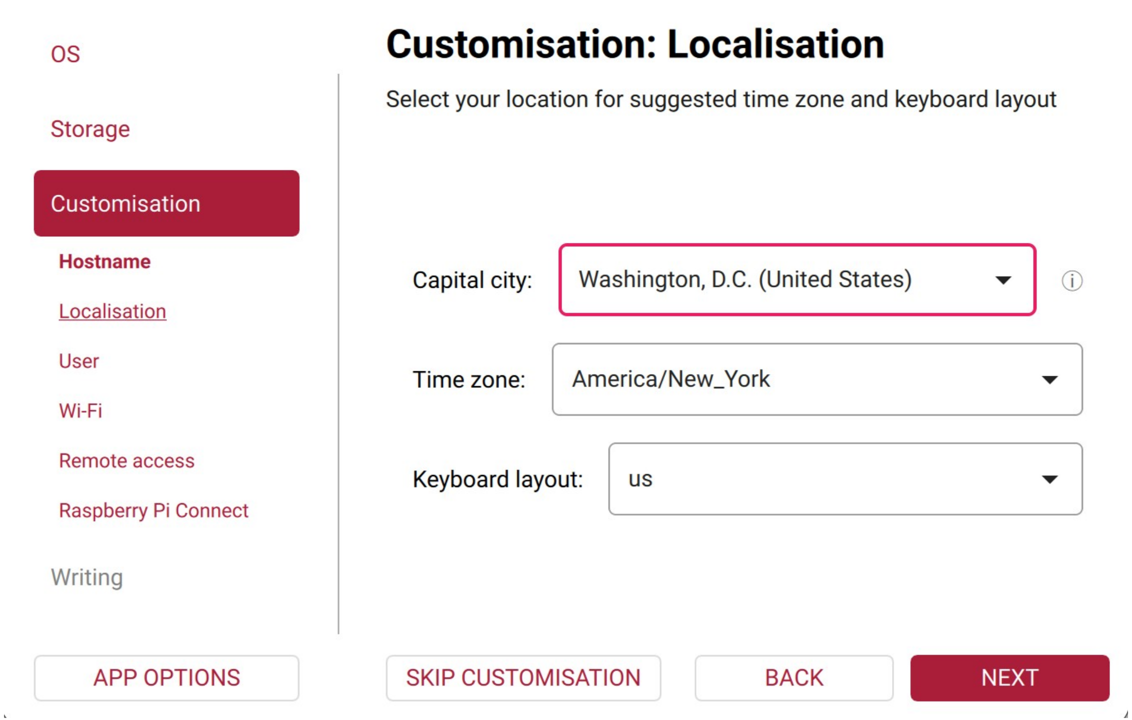
6



Open Raspberry Pi Imager

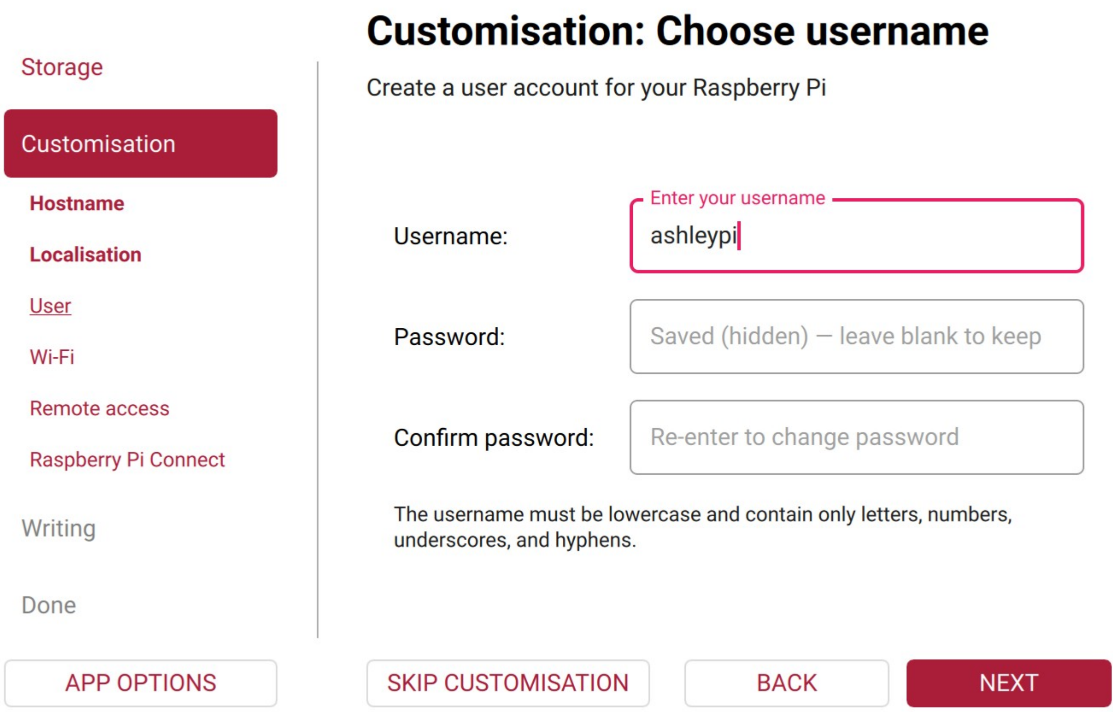
Customisation → Localisation

7



Open Raspberry Pi Imager
Customisation → User → Choose username

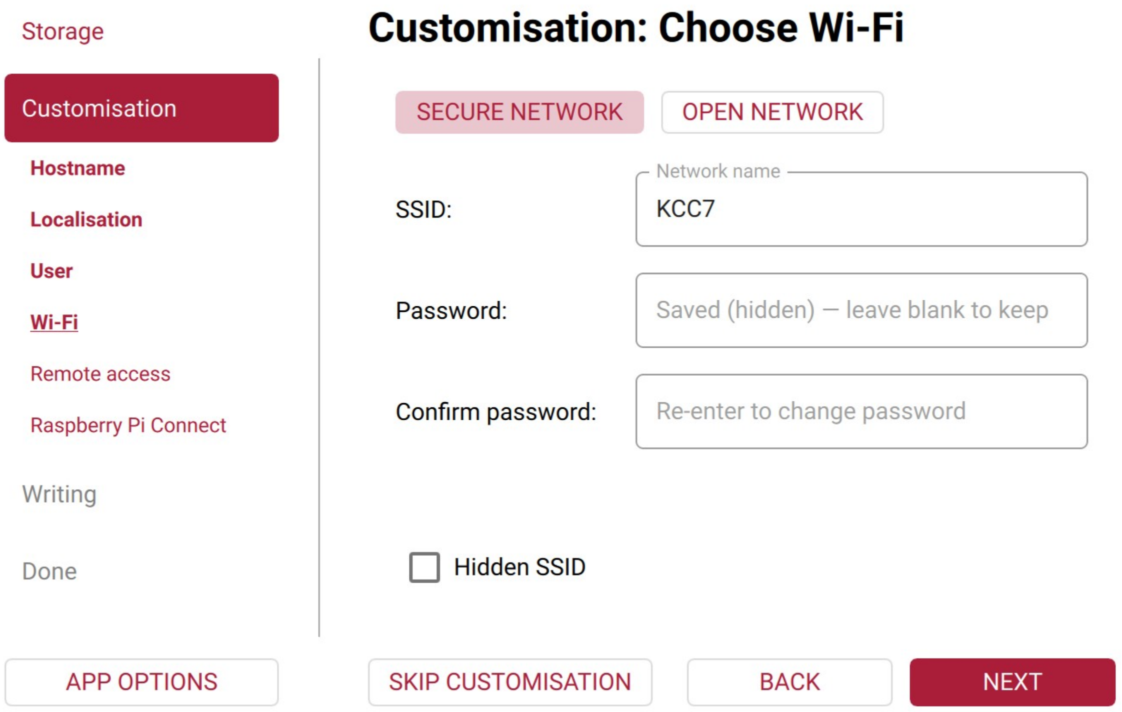
8



Open Raspberry Pi Imager

Customisation → Wi-Fi → Enter your Wi-Fi

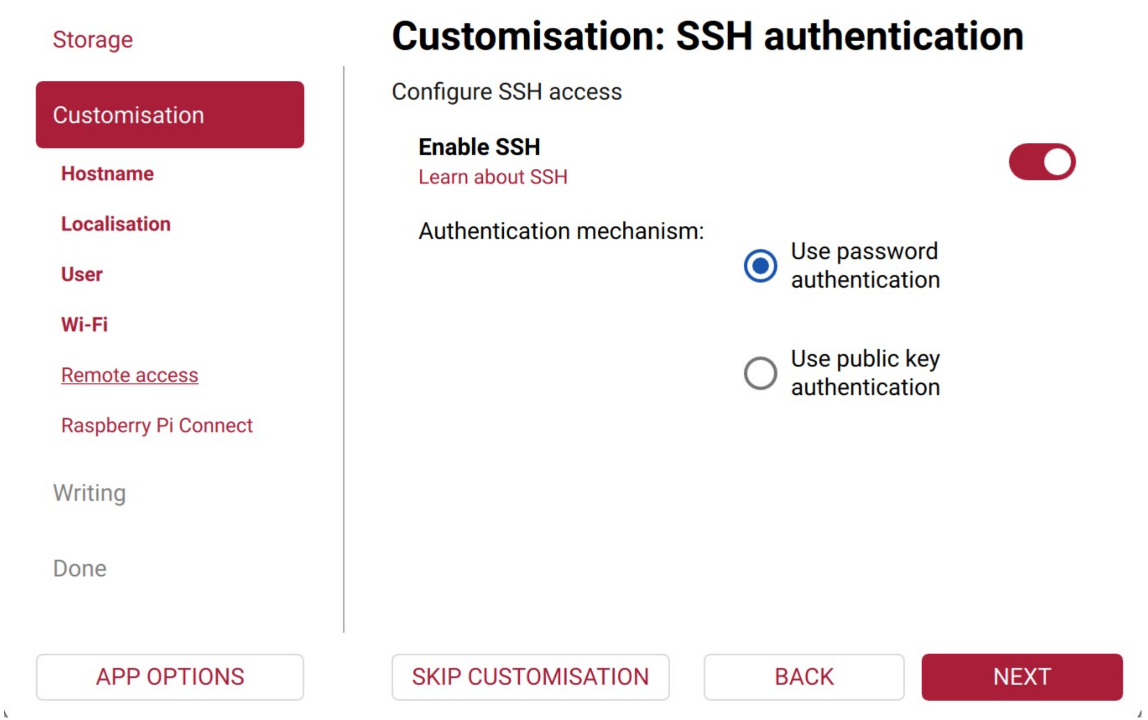
9



Open Raspberry Pi Imager

Customisation → SSH authentication → Use password authentication

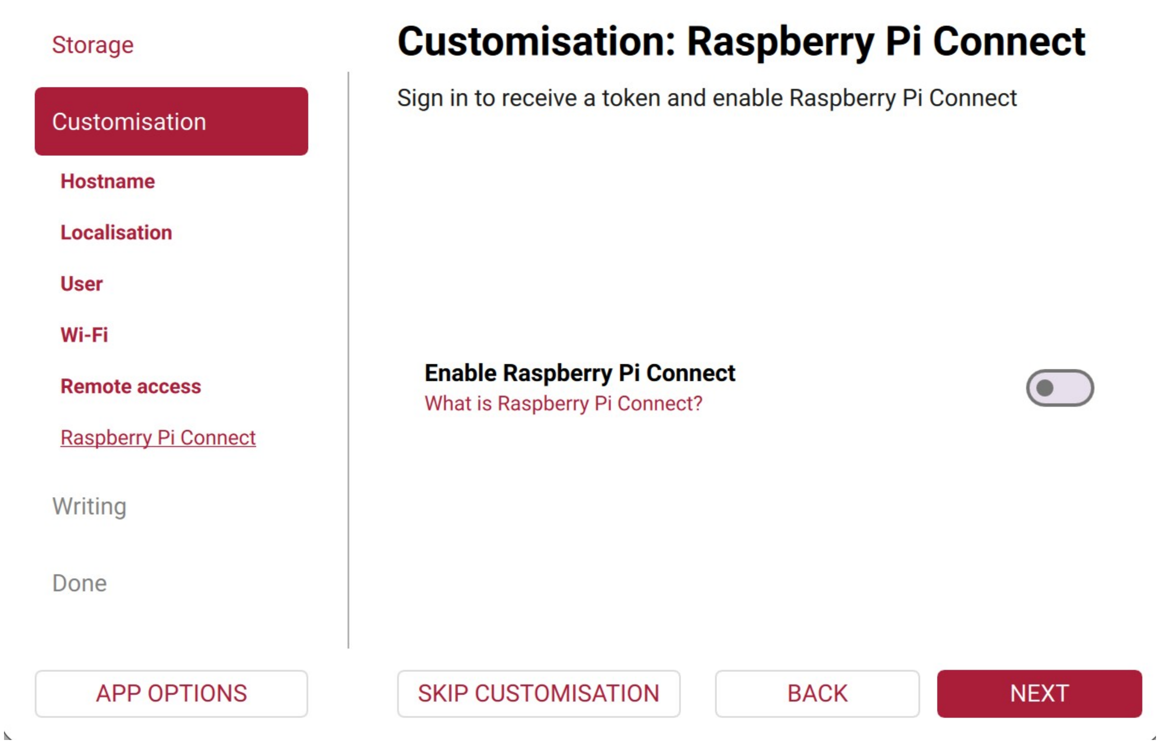
10



Open Raspberry Pi Imager

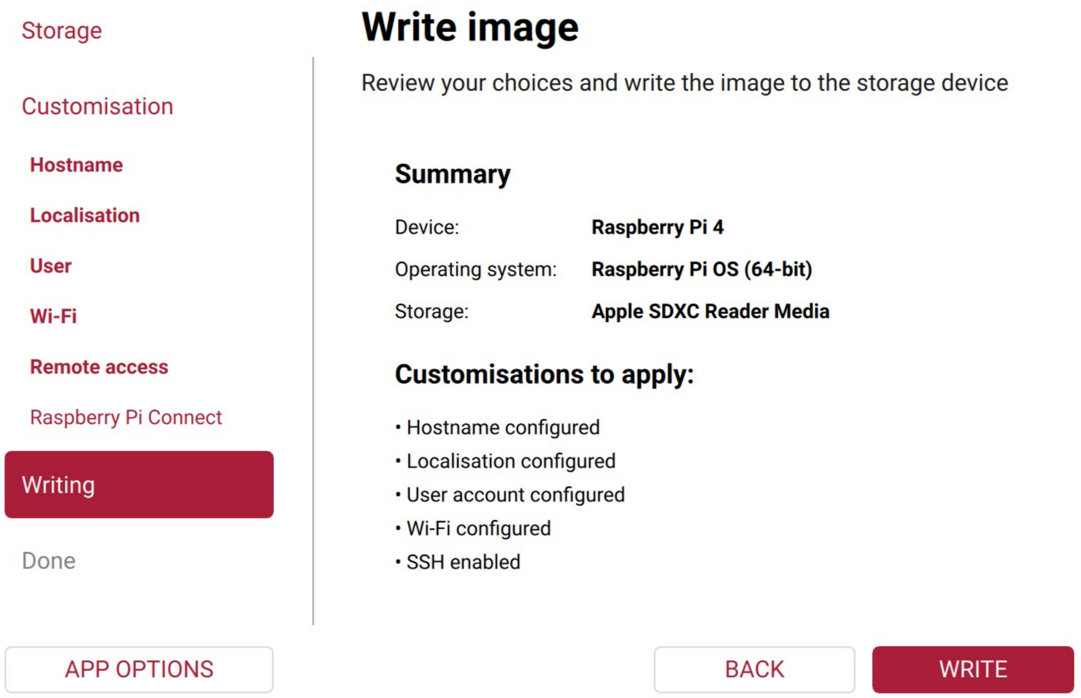
Customisation → Raspberry Pi Connect → Off

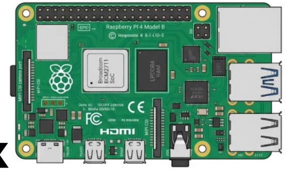
11

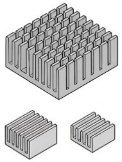


Open Raspberry Pi Imager
Writing → Write

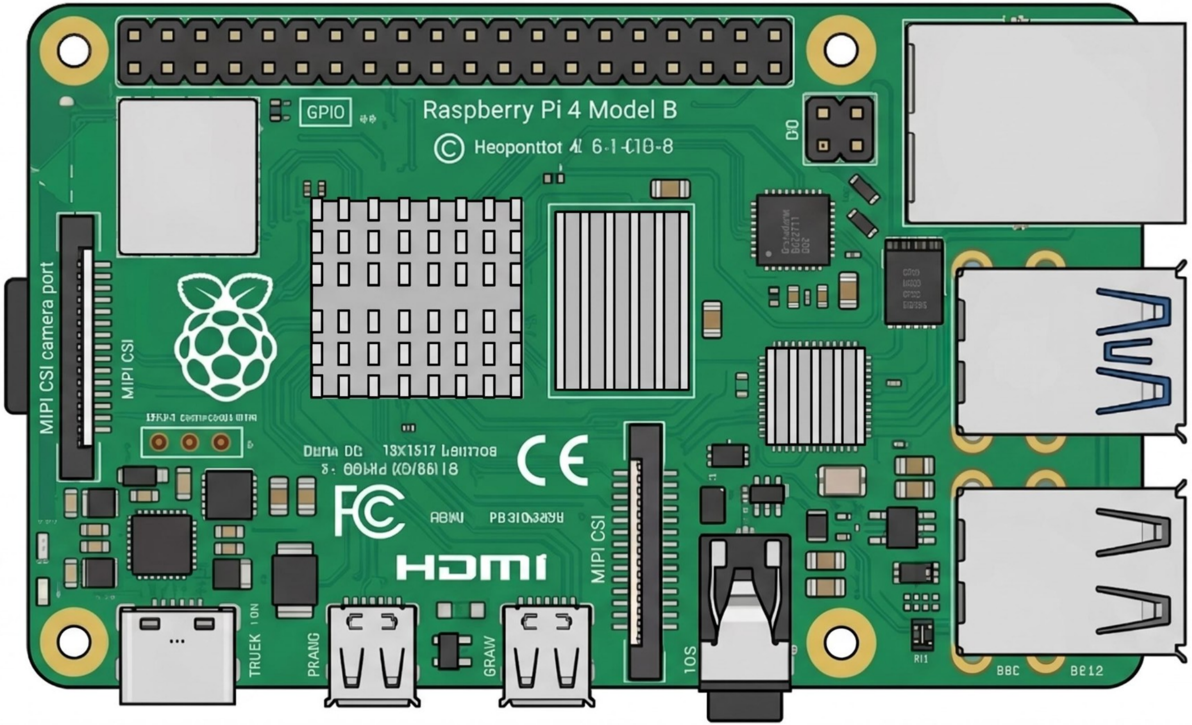
12

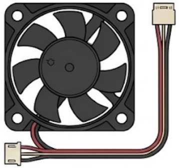


1x 

1x 

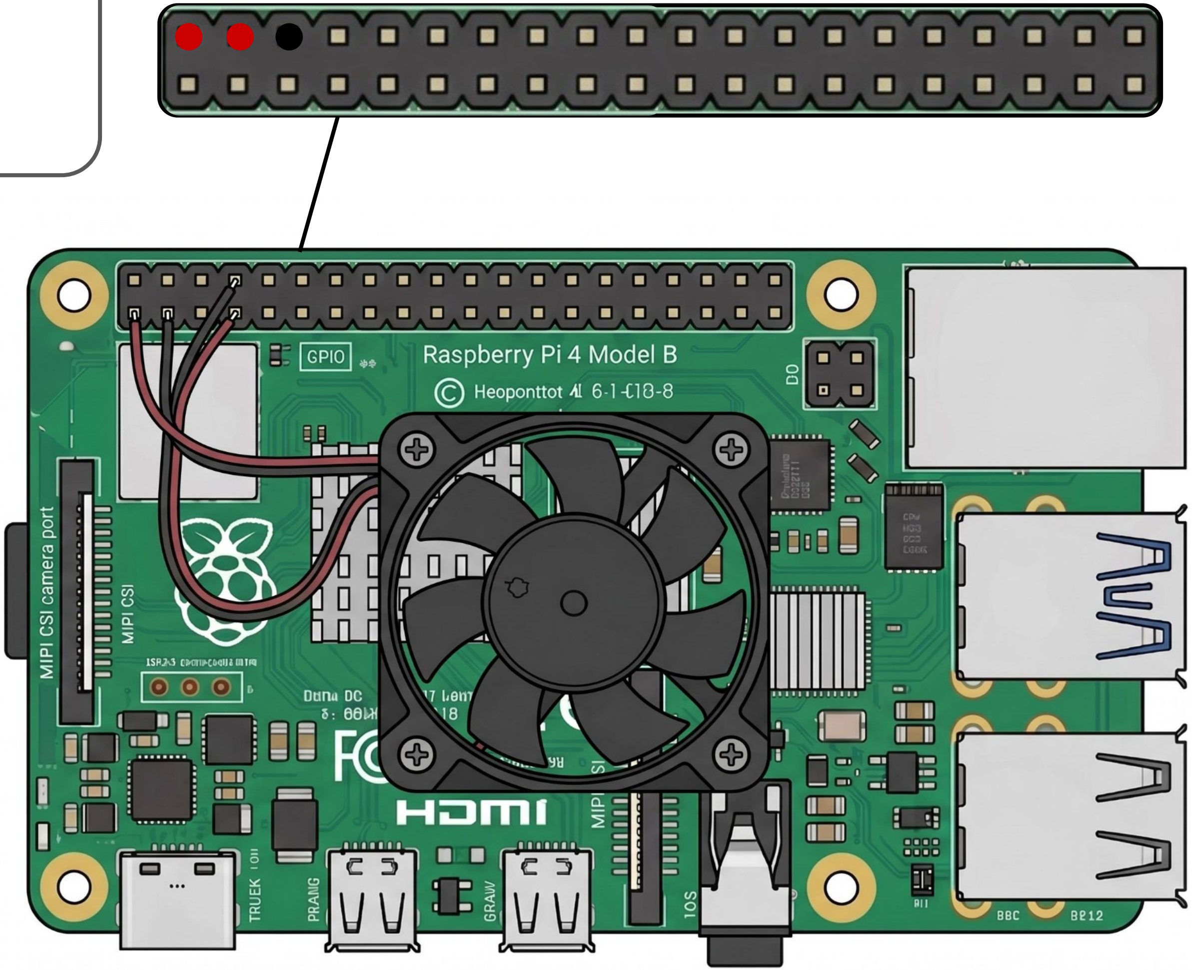
13


GPIO
Raspberry Pi 4 Model B
MIPI CSI camera port
MIPI CSI
HDMI
MIPI CSI



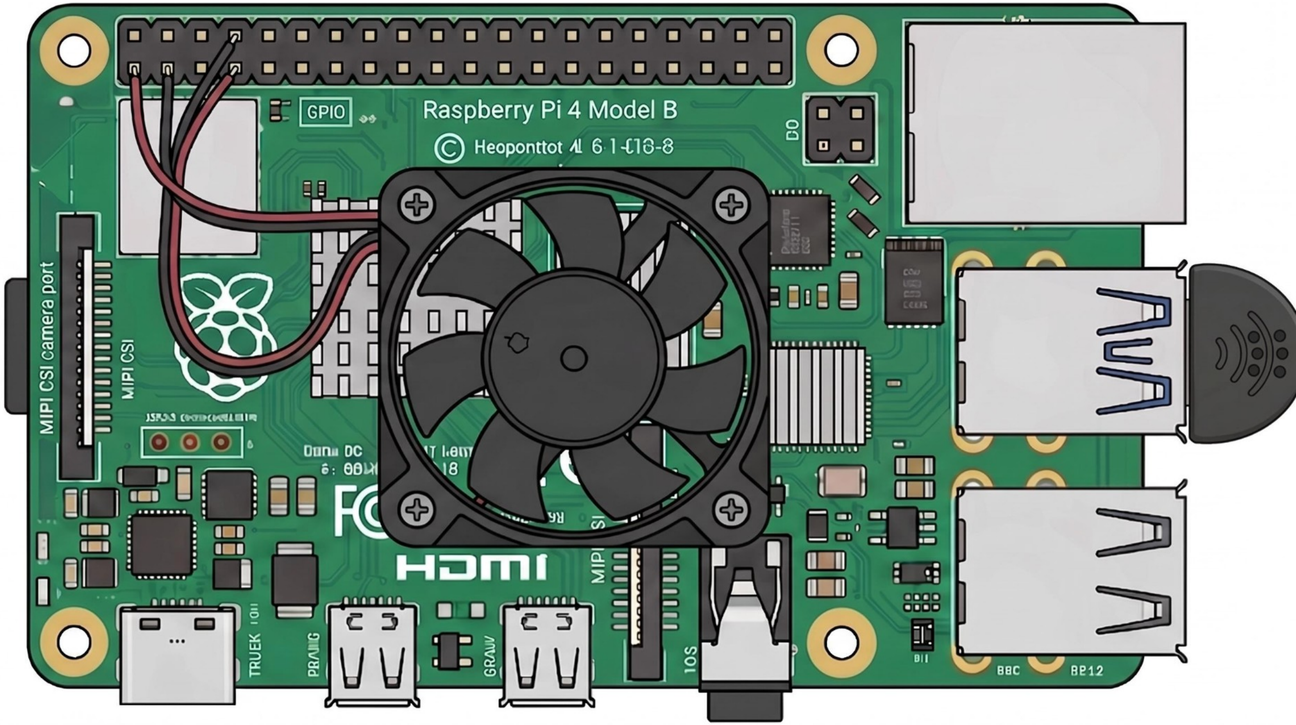
1x

14


GPIO
Raspberry Pi 4 Model B
MIPI CSI camera port
MIPI CSI
HDMI

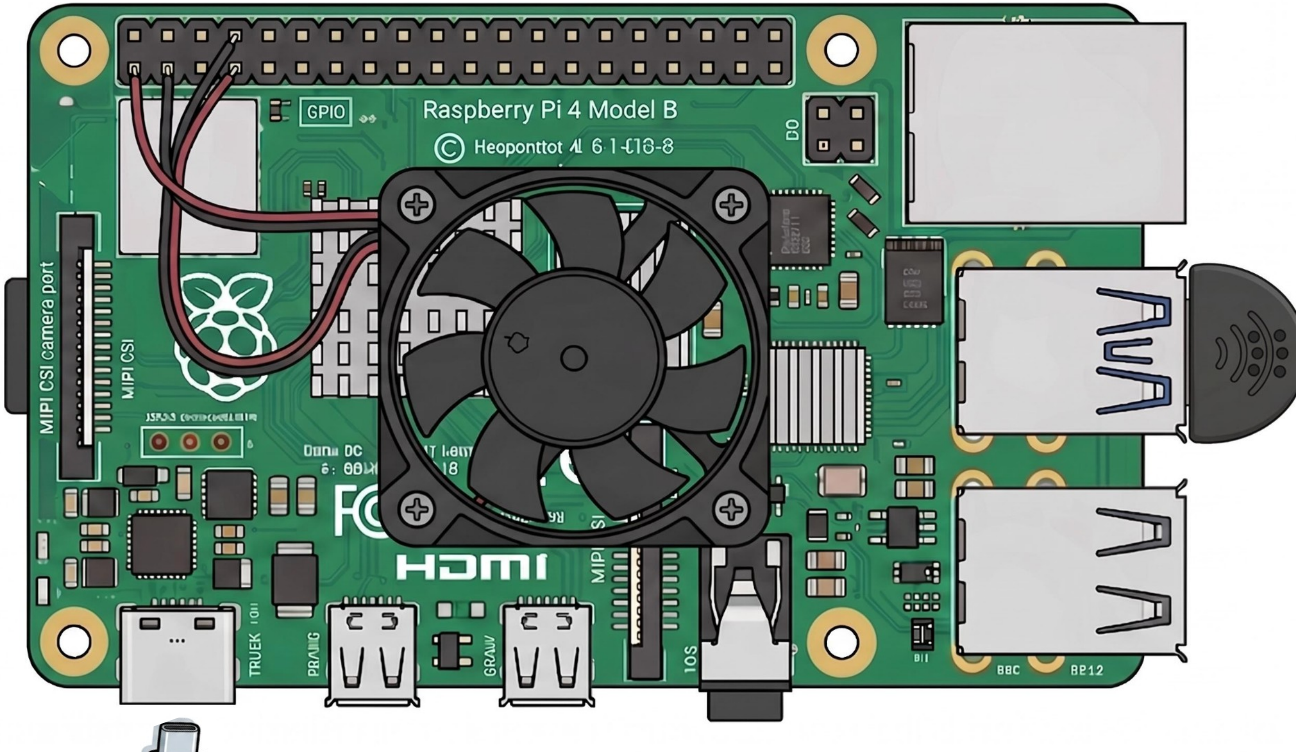
1x

15


Raspberry Pi 4 Model B
GPIO
MIPI CSI camera port
MIPI CSI
HDMI

1x 

16


GPIO
Raspberry Pi 4 Model B
MIPI CSI camera port
MIPI CSI
HDMI

Open Terminal on Mac

Connect to your Pi → ssh {PiName}@aslhand.local

17

```
(base) yourName@yourLaptopName ~ %
ashleypi@aslhand.local
```

Open Terminal on Mac

Run these commands

18

```
ashleypi@aslhand.local:~ $ sudo apt update
ashleypi@aslhand.local:~ $ sudo apt upgrade
-y
ashleypi@aslhand.local:~ $ sudo apt install
-y python3-pip i2c-tools portaudio19-dev
```

Open Terminal on Mac

Run this command then “Interface Options” → “I2C” → Enable

19

```
ashleypi@aslhand.local:~ $ sudo raspi-config
```

Open Terminal on Mac

Run this command

20

```
ashleypi@aslhand.local:~ $ pip3 install --break-system-packages adafruit-circuitpython-pca9685 adafruit-circuitpython-servokit SpeechRecognition pyaudio
```

Open Terminal on Mac

Test the microphone

The last line should output ~860K if the mic is working

21

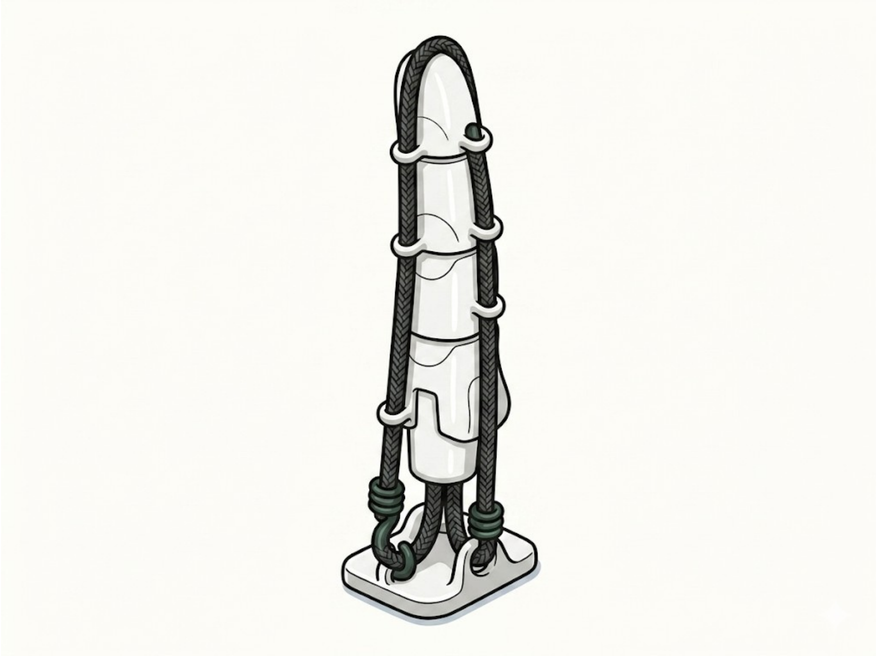
```
ashleypi@aslhand.local:~ $ arecord -l
ashleypi@aslhand.local:~ $ arecord -D plughw:1,0
-d 5 test.wav
ashleypi@aslhand.local:~ $ ls -lh test.wav
```

Par 

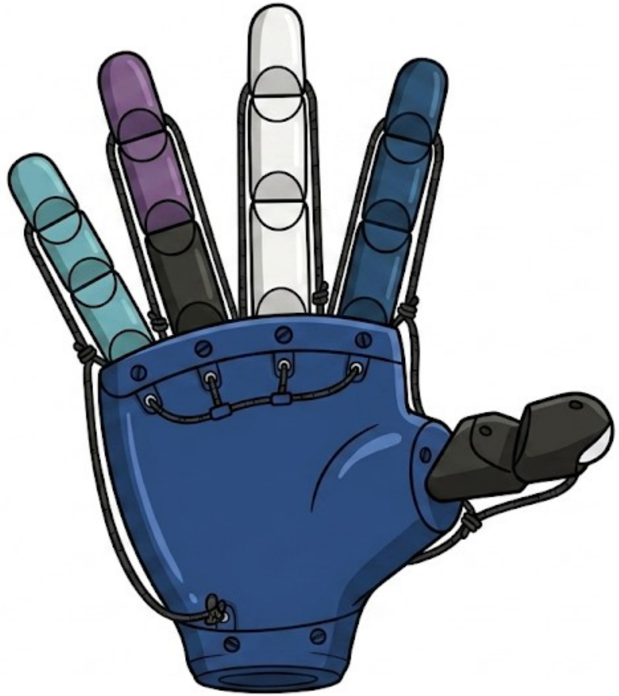
Set Up the Hand

5x  1x 

1

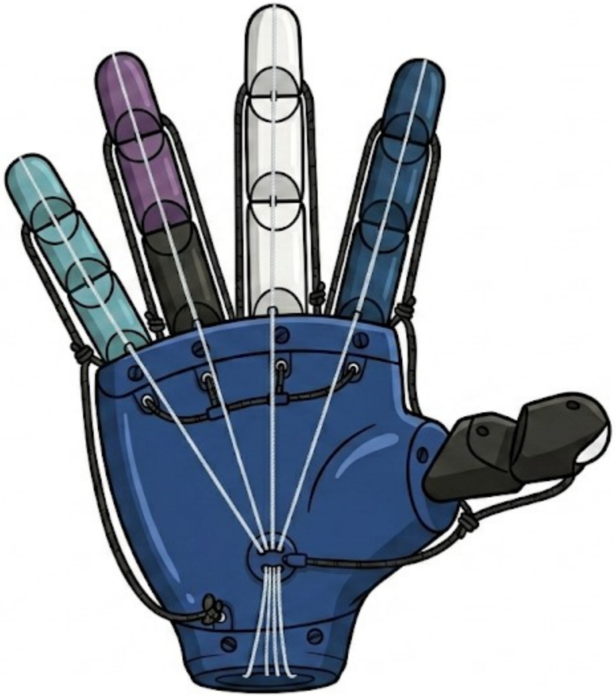


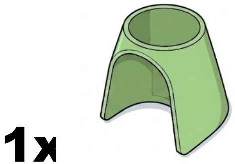
2



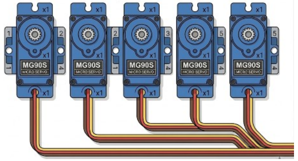
1x 

3

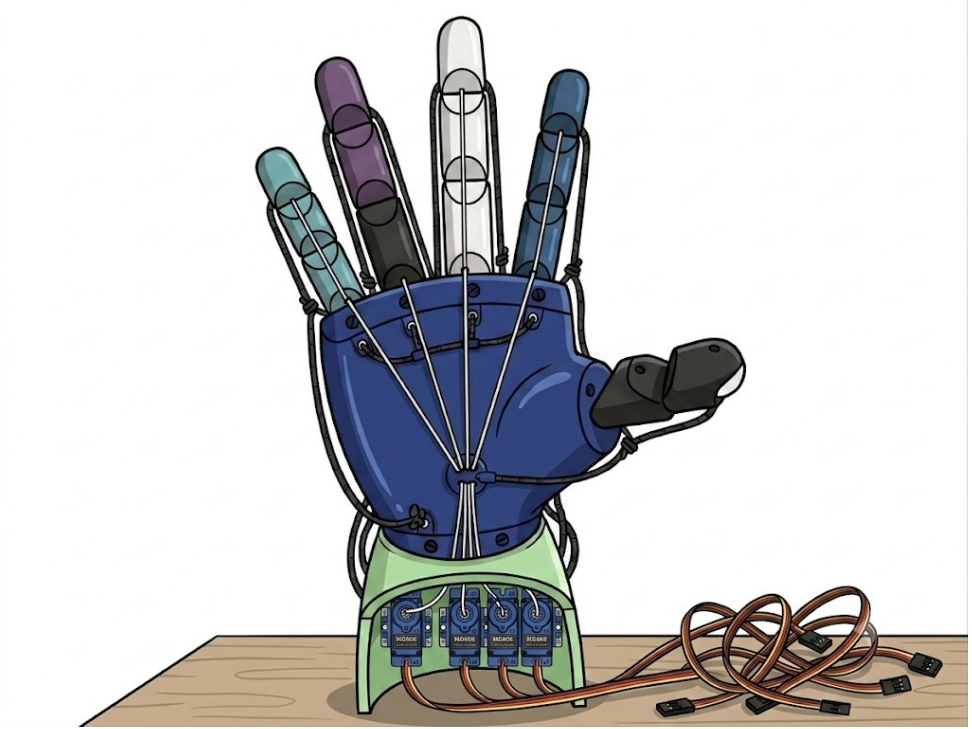




4

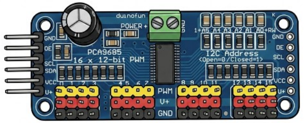
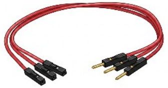
5x 

5



Part 

Connecting the Pi to the Hand

1x 
4x 

1

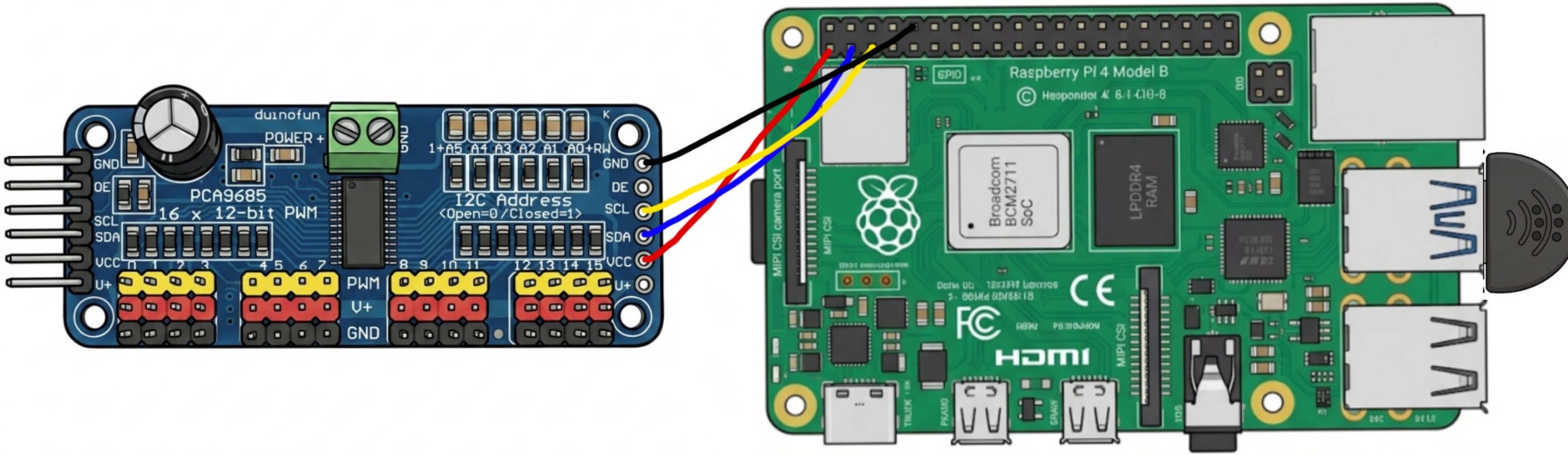
Raspberry Pi → PCA9685

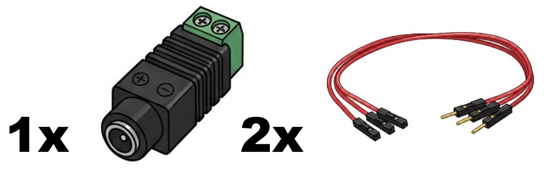
Pin 1 (3.3V) → VCC

Pin 9 (GND) → GND

Pin 3 (SDA GPIO 2) → SDA

Pin 5 (SCL GPIO 3) → SCL



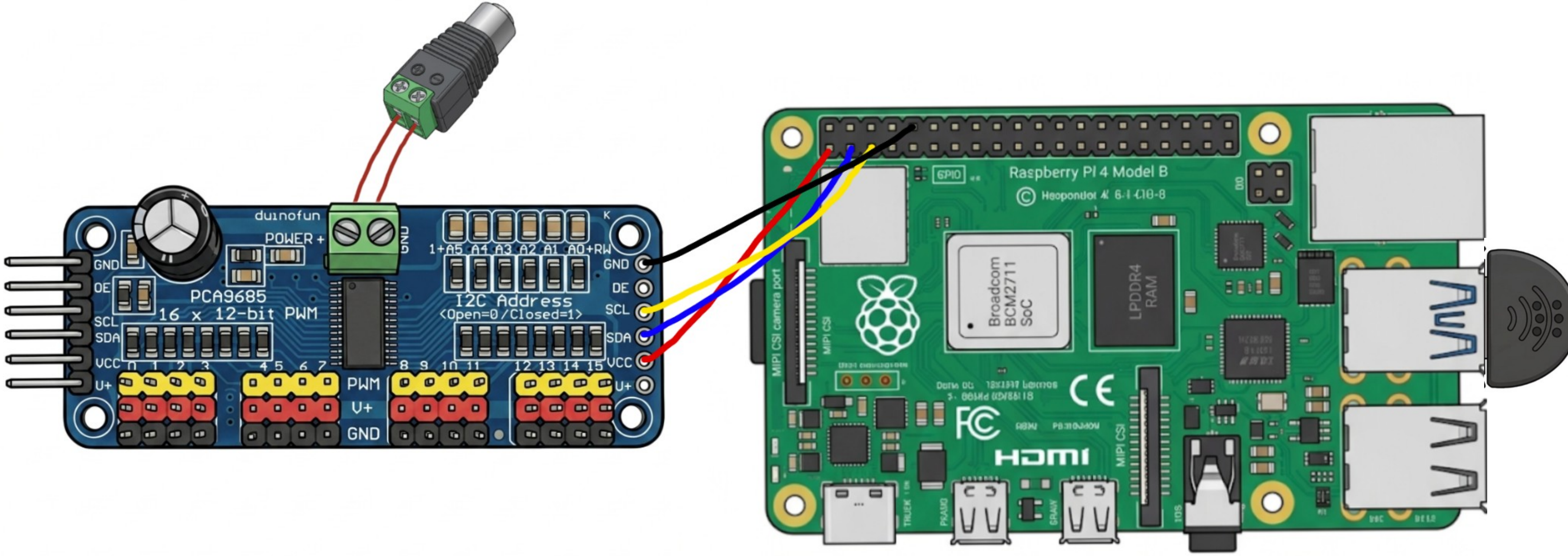


Barrel Jack Adapter → PCA9685

\+ → V+

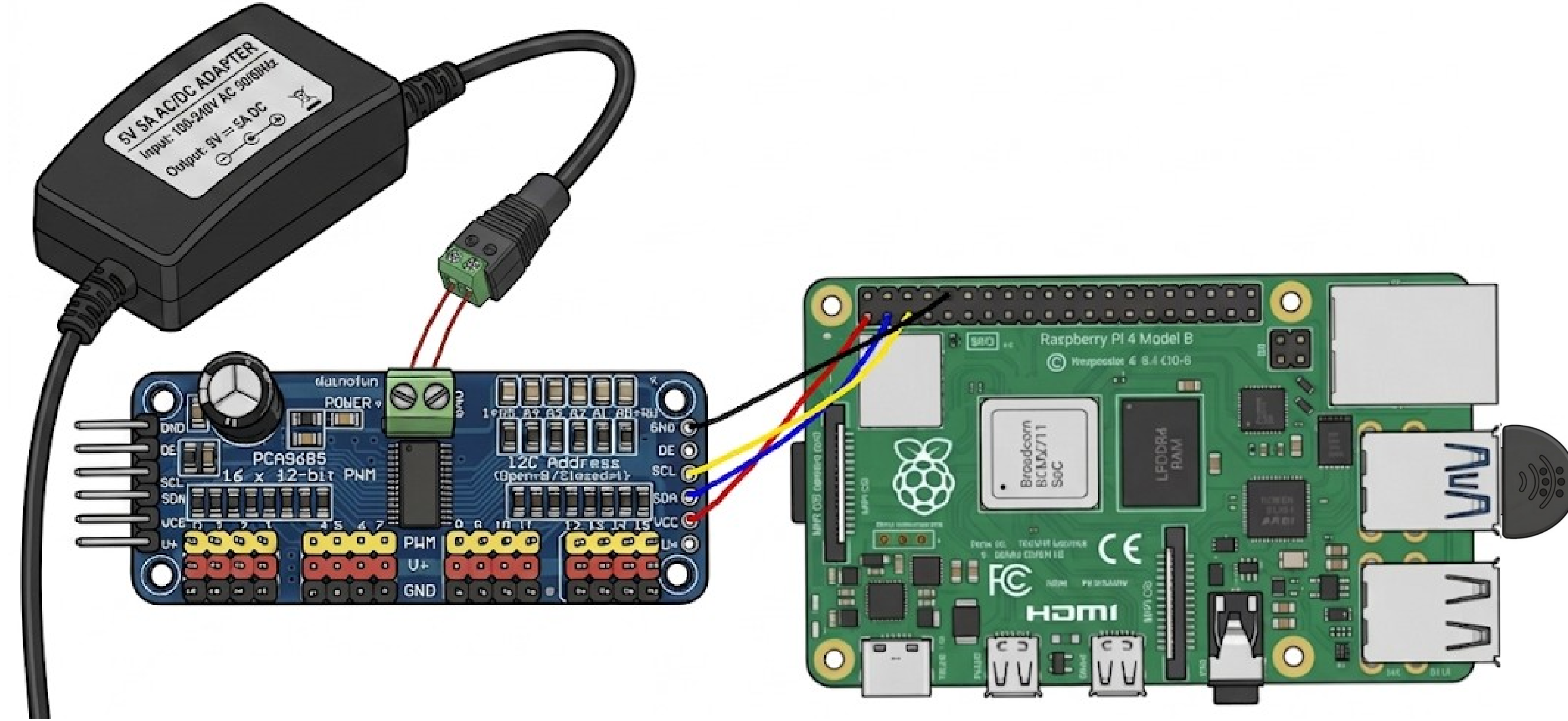
\- → GND

2



1x 

3


5V 5A AC/DC ADAPTER
Input: 100-240V AC 50/60Hz
Output: 5V ⎓ 5A DC
POWER
PCA9685
16 x 12-bit PWM
I2C Address
GND
SCL
SDA
VCC
PWM
V+
GND
HDMI

4

MG90S → PCA9685

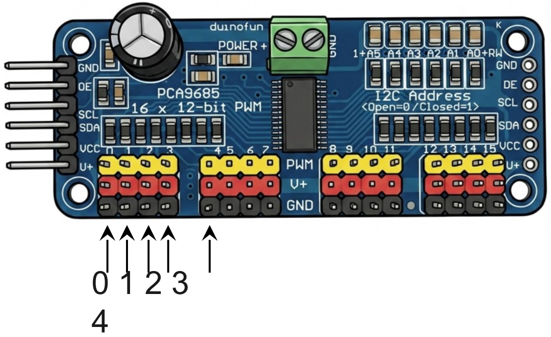
Thumb → Port 0

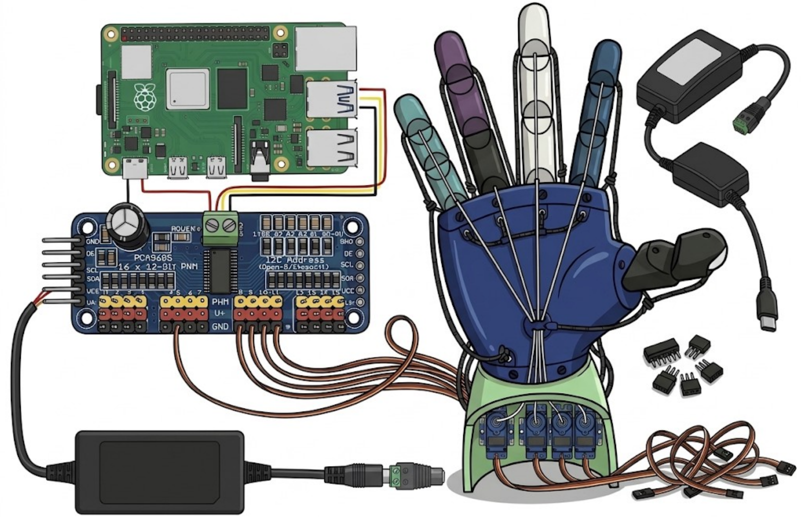
Index → Port 1

Middle → Port 2

Ring → Port 3

Pinky → Port 4





Make sure the yellow wire goes in the top row and the brown wire goes in the bottom row

Open Terminal on Mac
Run these commands

5

```
ashleypi@aslhand.local:~ $ nano
asl_hand.py
```

Open Terminal on Mac
Paste this script then Ctrl+O to save, Enter, then Ctrl+X to exit

6

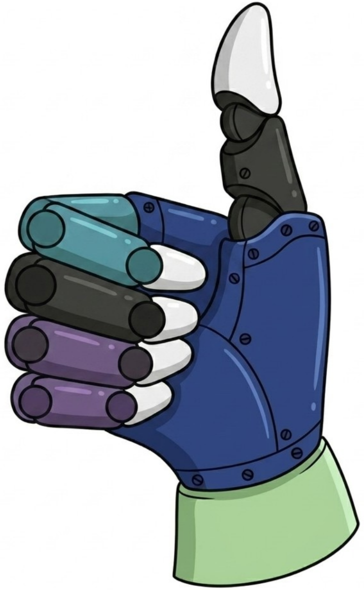
```
https://tinyurl.com/
5awuwn2y
```

Open Terminal on Mac

Run these commands

7

```
ashleypi@aslhand.local:~ $ python3
asl_hand.py
```



YOU’RE DONE!